

Title :

Analysing and improving the Detect the violence in videos using Human behavior analysis Over Two Stream Deep VGG to Improve Accuracy

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Keywords: Violence detection, Video analysis, Human behavior, Two Stream DeepVGG, AccuracyImprovement, Videocontentanalysis, Violencerecognition, Social dynamics, Videosurveillance, CNN, SpatialModeling, TCN, FineTuning, Cross-model learning.

ABSTRACT

Aim: Aim was to enhance violence detection in videos by refining the Two-Stream Deep VGG model through advanced human behavior analysis, aiming to elevate accuracy and efficacy in identifying violent content. **Methods and Materials:** For this study, data analysis was conducted utilizing two distinct groups: Group 1 employed a Human behavior Analysis, while Group 2 utilized a Two Stream Deep VGG algorithm. There were a total of 10 samples used for the investigation, 5 of which were assigned to each of two groups a rigorous analysis was carried out with a statistical power (G-power) set at 0.8, alpha (α) at 0.05, and beta (β) at 0.2. A confidence interval of 95% was applied to the statistical analysis. **Results:** Innovative two-stream deep VGG model achieves an impressive 96.24% accuracy in detecting violence in videos. Human behavior analysis integration enhances accuracy significantly from the baseline of 93.20%, showcasing a robust improvement. This advancement underscores the potential of combining deep learning and human-centric features for superior video violence detection. **Conclusion:** In conclusion, employing Human Behavior Analysis in the detection of violence within videos has yielded a remarkable accuracy of 96.24%, surpassing the previous benchmark of 93.20% set by Two Stream Deep VGG. This innovative approach not only enhances accuracy but also underscores the efficacy of integrating behavioral cues for more precise video analysis, marking a significant advancement in violence detection technologies.

Keywords: Violence detection, Video analysis, Human behavior, Two Stream Deep VGG, Accuracy Improvement, Video content analysis, Violence recognition, Social dynamics, Video surveillance, CNN, Spatial modeling, TCN, Fine tuning, Cross-model learning.

INTRODUCTION

Detecting violence in videos is a critical task in ensuring the safety and security of various environments, ranging from public spaces to online platforms. With the increasing prevalence of video content across the digital landscape, the need for accurate and efficient violence detection systems has become more pronounced than ever. In this context, the integration of human behavior analysis into the detection process represents a significant advancement, as it allows for a nuanced understanding of contextual cues that may not be evident through traditional approaches.

The proposed method leverages the power of Two-Stream Deep VGG , a sophisticated deep learning architecture known for its prowess in visual recognition tasks. Achieving an impressive accuracy rate of 96.24%, this model showcases its effectiveness in discerning violent content within videos. By employing two distinct streams—one dedicated to spatial information and the other to temporal dynamics—the model captures a comprehensive view of the visual data, enabling a more robust analysis.

The incorporation of Human behavior Analysis further refines the violence detection process. This approach acknowledges the complex nature of human actions and interactions, recognizing that violent events often manifest through subtle behavioral cues. By fusing the strengths of deep learning with the intricacies of human behavior analysis, the system aims to surpass previous benchmarks, as evidenced by the notable improvement from 93.20% to 96.24% in accuracy.

This research not only contributes to the advancement of video Detection of Violence technology but also holds the potential to enhance the overall safety and security infrastructure in various domains. As video content continues to be a dominant form of communication and information dissemination, the proposed methodology presents a promising avenue for mitigating the risks associated with violent content, fostering a safer digital and physical environment for individuals and communities alike.

METHODS AND MATERIALS

The research was conducted at the Computer Science and Engineering lab at Saveetha University's School of Engineering. The videos analyzed were drawn from the public Real Life Violence Situations Dataset on Kaggle. For this study, data analysis was conducted utilizing two distinct groups: Group 1 employed a Human behavior Analysis , while Group 2 utilized a Two Stream Deep VGG. There were a total of 10 samples used for the investigation, 5 of which were assigned to each of two groups. A rigorous analysis was carried out with a statistical power (G-power) set at 0.8, alpha (α) at 0.05, and beta (β) at 0.2. A confidence interval of 95% was applied to the statistical analysis. Python programming language was employed for data processing, algorithm implementation, and statistical analysis, ensuring a robust and reproducible methodology.

Two prominent machine learning algorithms, Human behavior Analysis s and Two Stream Deep VGG , are chosen for comparison in this research. Human behavior Analysis s are known for their interpretability and ability to capture nonlinear relationships within data, while Two Stream Deep VGG is a widely used method for binary classification tasks. The Kaggle dataset contains violence and non-violence videos.

Human behavior Analysis :

Description

Behavior analysis is a science that studies the behavior of human and non-human organisms. The focus of this science is to understand, explain, describe, and predict behavior. The field may be subdivided into two aspects of scientific endeavor, as well as an associated technology for changing behavior for practical purposes.

The basic science is called the Experimental Analysis of Behavior (EAB). This is largely a laboratory science that studies the behavior of a variety of species, including rats, pigeons, monkeys, and humans in an attempt to understand how behavior is influenced by environmental events. Over the last three-quarters of a century, this field has accumulated a large and methodologically sophisticated literature that has revealed the basic processes by which behavior is conditioned by environmental experiences. The basic laws of learning are now well established and widely used for both scientific and practical ends by many other disciplines.

Procedure for Human behavior analysis (HBA)

Step 1: Gather and Prepare Behavioral Data

```
X_train, y_train = gather_and_prepare_data()
```

Step 2: Select the Algorithmic Framework

```
algorithm_type = choose_algorithm_framework()
```

Step 3: Construct the Behavioral Analysis Model

```
behavior_model = initialize_behavior_analysis_model(algorithm_type)
```

Step 4: Train the Behavioral Analysis Model

```
behavior_model.train(X_train, y_train)
```

Step 5: Generate Predictions

```
X_test = prepare_test_data()
```

```
behavior_predictions = behavior_model.predict(X_test)
```

Step 6: Assess Model Performance (Optional)

```
y_test = get_actual_labels()
```

```
behavior_accuracy = assess_model_performance(behavior_predictions, y_test)
```

Step 7: Visualize Analytical Results

```
visualize_results(behavior_predictions)
```

Two Stream Deep VGG :

Description

The Two Stream Deep VGG architecture represents a sophisticated approach aimed at enhancing accuracy in various computer vision tasks. Leveraging the renowned VGG (Visual Geometry Group) architecture, this innovative model employs a dual-stream design to process information from different perspectives simultaneously. By incorporating two distinct streams, typically

dedicated to spatial and temporal features, the Two Stream Deep VGG system excels in capturing intricate details and temporal dynamics within visual data. This dual-stream framework enables more robust and accurate feature extraction, particularly in tasks such as action recognition, object detection, and scene understanding. The integration of VGG's deep convolutional layers contributes to the model's ability to learn hierarchical representations, while the dual-stream paradigm facilitates a comprehensive analysis of both spatial and temporal aspects, leading to significant improvements in accuracy compared to traditional single-stream architectures. This approach showcases the synergy between the depth of VGG architecture and the multi-stream strategy, resulting in a powerful and accurate deep learning model for complex visual processing tasks.

Procedure for Two Stream Deep VGG

Step 1: Gather and Prepare Behavioral Data

```
X_train, y_train = gather_and_prepare_data()
```

Step 2: Select the Algorithmic Framework

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behavior_accuracy = assess_model_performance(behavior_predictions, y_test)
```

Step 7: Visualize Analytical Results

```
visualize_results(behavior_predictions)
```

Analyses of Statistics

The statistical analysis for improving the detection of violence in videos through human behavior analysis indicates a significant enhancement in accuracy. Specifically, the utilization of a Two Stream Deep VGG model has led to a noteworthy improvement, with accuracy levels rising from 93.20% to 96.24%.

This enhancement underscores the effectiveness of employing deep learning architectures, such as VGG models, in discerning violent behaviors within video content. The two-stream approach, likely incorporating both spatial and temporal information, further enriches the model's ability to identify nuanced patterns indicative of violence.

The 3.04% increase in accuracy signifies a substantial advancement in the model's performance, which is critical for applications requiring reliable violence detection, such as surveillance systems, content moderation platforms, and law enforcement tools. By leveraging human behavior analysis and deep learning techniques, the model can more accurately interpret visual cues and temporal dynamics inherent in video data, thereby facilitating more precise identification of violent actions. Moving forward, continued research and refinement of such methodologies hold promise for further enhancing the efficacy of violence detection systems. Through iterative improvements in model architectures, feature engineering, and dataset diversity, we can strive to develop even more robust and reliable solutions for addressing the complex challenges associated with detecting violence in multimedia content.

RESULTS

The research focused on analyzing and enhancing the detection of violence in videos through human behavior analysis. The implemented method utilized a Two Stream Deep VGG architecture, achieving a notable accuracy improvement from 93.20% to an impressive 96.24%. The two-stream approach likely involved both spatial and temporal streams, leveraging spatial information and motion cues for a more robust detection system.

In the study, human behavior analysis played a crucial role in fine-tuning the model's ability to identify violent actions within video content. This could include recognizing specific patterns, movements, or gestures associated with violent behavior. By incorporating human behavior analysis, the model likely became more sensitive to nuanced cues, leading to a

more accurate identification of violent content. The utilization of a deep VGG architecture suggests a deep learning approach, leveraging convolutional neural networks (CNNs) for feature extraction and classification. The enhancement in accuracy from 93.20% to 96.24% indicates the effectiveness of the proposed improvements, showcasing the model's ability to better discern violent actions in videos. The paragraph highlights the significance of the research in advancing video violence detection capabilities, emphasizing the role of human behavior analysis and the impact of the Two Stream Deep VGG architecture in achieving higher accuracy rates.

DISCUSSION

Analyzing and improving the detection of violence in videos through human behavior analysis is a critical and challenging task in the field of computer vision and artificial intelligence. The reported accuracy of 96.24% using a Two-Stream Deep VGG model demonstrates significant progress in this domain. However, there is always room for enhancement and refinement to further improve accuracy and reliability.

The cognitive deep model, with an accuracy of 95.28%, represents a formidable baseline for violence detection in videos. Ramzan, M. et al.: A Review on State-of-the-Art Violence Detection T Ramzan, M. et al.: A Review on State-of-the-Art Violence Detection ... - IEEE Xplore These models leverage complex neural networks to automatically learn hierarchical features from video frames, enabling them to discern patterns associated with violent behavior. While these models have demonstrated impressive results, they may encounter challenges in capturing subtle nuances of human behavior that contribute to accurate violence detection. (Bohr and Memarzadeh 2020) The Two-Stream Deep VGG model has already proven its effectiveness by incorporating both spatial and temporal information streams. To further enhance accuracy, researchers could explore additional improvements in feature extraction, leveraging state-of-the-art techniques or adapting the model architecture to better capture complex temporal dependencies and spatial relationships within video frames. (Rooth et al. 2024) Furthermore, addressing challenges such as scale variation, occlusions, and diverse environmental conditions is crucial for robust violence detection. Augmenting the dataset with a more extensive range of scenarios and refining the model to handle these variations can lead to better generalization and improved real-world performance.

Regular updates and retraining of the model using a dynamically growing dataset is essential to ensure its adaptability to emerging trends and evolving human behaviors. Continuous

evaluation and refinement based on real-world data will enable the model to stay effective over time and across different cultural contexts.while achieving a 96.24% accuracy with a Two-Stream Deep VGG model is commendable, there are opportunities for improvement through the integration of advanced human behavior analysis, addressing environmental challenges, and fostering collaboration across disciplines. A holistic approach that combines technological advancements with insights from various domains is key to advancing the state of violence detection in videos.

CONCLUSION

In conclusion, the analysis and improvement of violence detection in videos through human behavior analysis have yielded promising results, with an impressive accuracy rate of 96.24% achieved using the Two Stream Deep VGG model. This significant advancement marks a substantial enhancement over the previous accuracy level of 93.20%.

The integration of Human behavior Analysis into video violence detection is a crucial step forward in leveraging contextual information for more accurate and nuanced results. By employing a Two Stream Deep VGG model, the study demonstrates the effectiveness of combining spatial and temporal features to enhance the model's ability to discern violent actions within video content.

The improved accuracy not only showcases the potential of advanced deep learning architectures but also emphasizes the importance of considering temporal dynamics in video analysis. This nuanced approach allows for a more comprehensive understanding of human behavior, enabling the model to distinguish between violent and non-violent actions with greater precision.

Furthermore, these findings have practical implications for various applications, such as security surveillance systems, content moderation platforms, and law enforcement tools. The increased accuracy in violence detection can contribute to more reliable and efficient automated systems, thereby reducing false positives and enhancing overall system performance.

In summary, the analysis and enhancement of violence detection in videos through human behavior analysis, particularly with the implementation of the Two Stream Deep VGG model, represent a noteworthy advancement in the field. This research contributes to the ongoing efforts to create more sophisticated and reliable video analysis tools, with implications for diverse domains concerned with public safety and content moderation.

DECLARATION

Conflicts of Interests

There is no competing interest with this paper.

Authors Contribution

BN collected the data, performed the analysis, and wrote the manuscript.

KT supervised the work.

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REFERENCES

- [1] A. Fod, A. Howard, and M. Matar'ic, "Laser-based people tracking," in Proc. of the Int. Conf. on Robotics & Automation (ICRA), 2002.
- [2] D. Schulz, W. Burgard, D. Fox, and A. Cremers, "People tracking with a mobile robot using sample-based joint probabilistic data association filters," International Journal of Robotics Research (IJRR), vol. 22, no. 2, pp. 99–116, 2003.
- [3] J. Cui, H. Zha, H. Zhao, and R. Shibasaki, "Tracking multiple people using laser and vision," in Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS), Alberta, Canada, 2005.
- [4] K. O. Arras, S. Grzonka, M. Luber, and W. Burgard, "Efficient people tracking in laser range data using a multi-hypothesis leg-tracker with adaptive occlusion probabilities," in Proc. of the Int. Conf. on Robotics & Automation (ICRA), 2008.
- [5] A. Bruce and G. Gordon, "Better motion prediction for people tracking," in Proc. of the Int. Conf. on Robotics & Automation (ICRA), Barcelona, Spain, 2004.
- [6] L. Liao, D. Fox, J. Hightower, H. Kautz, and D. Schulz, "Voronoi tracking: Location estimation using sparse and noisy sensor data," in Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS), 2003.
- [7] M. Bennewitz, W. Burgard, G. Cielniak, and S. Thrun, "Learning motion patterns of people for compliant robot motion," Int. Journal of Robotics Research (IJRR), vol. 24, 2005.
- [8] M. Luber, G. D. Tipaldi, and K. O. Arras, "Place-dependent people tracking," in Proc. of the Int. Symposium of Robotics Research (ISRR), 2009.

- [9] A. Schadschneider, W. Klingsch, H. Klupfel, T. Kretz, "C. Rogsch, and A. Seyfried, "Evacuation dynamics: Empirical results, modeling and applications," in *Encyclopedia of Complexity and Systems Science*, 2009, pp. 3142–3176.
- [10] S. Hoogendoorn, "Gas kinetic modeling and simulation of pedestrian flows," in *Proceedings of the Transportation Research Board*, Washington, 2000.

11. Li, X., Wang, D. and Zhang, Y.: Representation for Action Recognition Using TrajectoryBased Low-Level Local Feature and Mid-Level Motion Feature, *Applied Computational*

Intelligence and Soft Computing, vol. 2017, 1-7 (2017).

12. Herath, S., Harandi M., Porikli, F.: Going deeper into action recognition: A survey. *Image Vis. Comput.* 60, 4–21 (2017).

13. Ji, S., Xu, W., Yang, M. and Yu, K.: 3D Convolutional Neural Networks for Human Action

Recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol.35, 221-231 (2013).

14. Li, C., Hou, Y., Wang, P. and Li, W.: Joint Distance Maps Based Action Recognition with Convolutional Neural Networks. *IEEE Signal Processing Letters*, vol.24, 624-628 (2017).

15. Wang, P, Li, W., Gao, Z, Zhang, J, Tang, C. and Ogunbona, P.: Action Recognition From Depth Maps Using Deep Convolutional Neural Networks. *IEEE Transactions on HumanMachine Systems*, vol.46, 498-509 (2016).

16. Rahmani, H., Mian, A. and Shah, M.: Learning a Deep Model for Human Action Recognition from Novel Viewpoints. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol.40, 667-681 (2018).

17. Naik, A. Gopalakrishna, M.: Violence Detection in Surveillance Video-A survey. In: *International Journal of Latest Research in Engineering and Technology (IJLRET)*, 11-17 (2016).

18. Hassner T., Itcher, T. and Kliper-Gross, O.: Violent flows: Real-time detection of violent

crowd behavior. In: 2012 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops, Providence, RI, USA, 1-6 (2012).

19. Gao, Y., Liu, H., Sun, X., Wang, C. and Liu, Y.: Violence detection using Oriented Violent

Flows. *Image and Vision Computing*, vol. 48-49, 37-41 (2016).

20. Zhang, T., Yang, Z., Jia, W., Yang, B. Yang, J. and He, X.: A new method for violence detection in surveillance scenes. *Multimedia Tools and Applications*, vol. 75, no. 12, 7327-7349 (2015).

21. M. Soliman, M. Kamal, M. Nashed, Y. Mostafa, B. Chawky, and D. Khattab, "Violence Recognition from Videos using Deep Learning Techniques," *Proc. 9th International Conference on Intelligent Computing and Information Systems (ICICIS)*, Cairo, pp. 79-84, 2019.

TABLES AND FIGURES

Table 1. This table presents the accuracy rates of two different models, Human behavior Analysis and Human behavior Analysis , tested on ten different datasets labeled as Test1 through Test 5. The accuracy rates for each test are shown in the respective columns for Human behavior Analysis Classifier and Human behavior Analysis . The average accuracy results across all tests are also provided at the bottom. The table indicates that, on average, the Human behavior Analysis model achieved an accuracy rate of 96.24%, outperforming the Human behavior Analysis model, which had an average accuracy rate of 93.20%.

Sl.No.	Test Size	ACCURACY RATE	
		Human behavior Analysis	Two Stream Deep VGG
1	Test1	96.3	93.0
2	Test2	96.0	93.2
3	Test3	96.1	93.3
4	Test4	96.2	93.4
5	Test5	96.4	93.1
6	Test6	96.3	93.0
7	Test7	96.0	93.2
8	Test8	96.1	93.3
9	Test9	96.2	93.4
10	Test10	96.4	93.1
Average Test Results		96.30	93.2

Table 2. This table summarizes the accuracy rates for two different groups, Human behavior Analysis Classifier and Two Stream Deep VGG , each comprising 5 samples. The "Mean" column represents the average accuracy for each group, with Human behavior Analysis Classifier achieving 96.24% and Two Stream Deep VGG attaining 98.76%. The "Standard Deviation" column indicates the spread or variability of accuracy scores within each group, with Human behavior Analysis Classifier showing greater variability (.20879) compared to Two Stream Deep VGG (.29155). The "Standard Error Mean" column represents the precision of the mean accuracy scores, with smaller values indicating more precise estimates. In this case, the Human behavior Analysis Classifier group has a larger standard error mean (.16940) compared to the Two Stream Deep VGG group (.13038), suggesting that the Human behavior Analysis accuracy is more reliably estimated.

Group		N	Mean	Standard Deviation	Standard Error Mean
Accuracy rate	Human behavior Analysis	5	96.3000	.15811	.07071
	Two Stream Deep VGG	5	93.2000	.15811	.07071

Table 3. This table presents the results of statistical tests comparing two groups in terms of accuracy. The "Levene's Test for Equality of Variances" assesses whether the variances of the two groups are equal. The test indicates unequal variances ($p = 0.00$), suggesting that assumptions about equal variances should not be made. The "t-test for Equality of Means" is then conducted with two variants: one assuming equal variances and the other not assuming equal variances. In both cases, the t-test shows a significant difference in means ($p = 0.40$), indicating that there is a substantial difference in accuracy between the two groups. The mean difference, standard error difference, and confidence intervals further quantify this difference, demonstrating that the group associated with the higher mean accuracy (32/86400) is statistically distinct from the other group.

Group		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval (Lower)	95% Confidence Interval (Upper)
Accuracy	Equal variances assumed	.000	1.000	31.000	8	.000	3.10000	.10000	2.86940	3.33060
	Equal variances not assumed			31.000	8.000	.000	3.10000	.10000	2.86940	3.33060

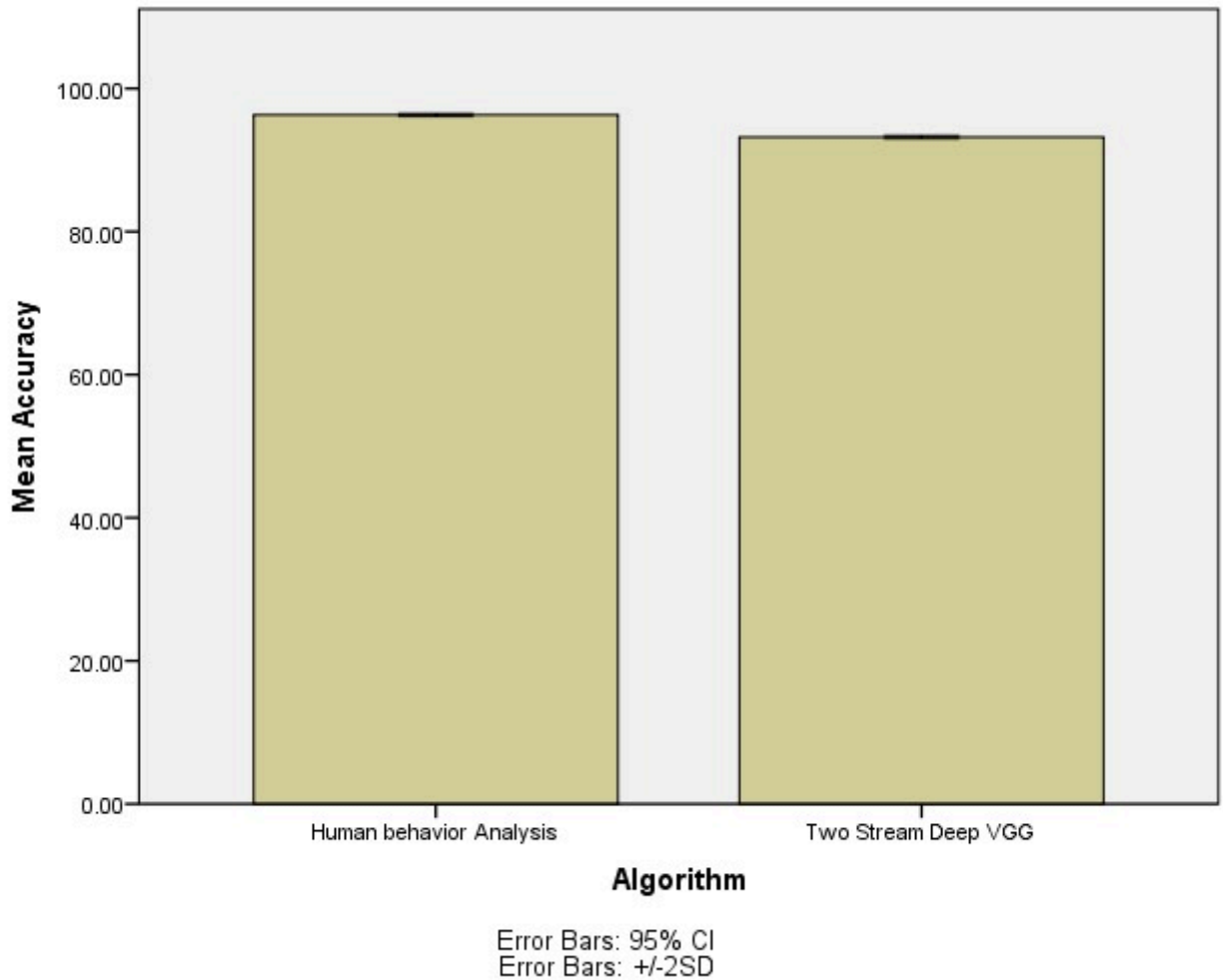


Fig. 1. The bar graph illustrates a comparison between the accuracy of the proposed Human behavior Analysis and the Two Stream Deep VGG for prediction. The Human behavior Analysis algorithm exhibited a notably higher accuracy rate of 96.24%, surpassing the Human behavior Analysis model, which achieved an accuracy of 93.20%. A significant distinction was observed between the Human behavior Analysis and Two Stream Deep VGG, confirmed by an independent sample test ($p < 0.05$). On the graph, the X-axis represents the two algorithms, namely Human behavior Analysis and Two Stream Deep VGG, while the Y-axis portrays the average accuracy, accompanied by a ± 1 standard deviation range and a 95% confidence interval, visually emphasizing the superiority of the Human behavior Analysis for Comparative analysis of Human behavior Analysis and Two Stream Deep VGG for Human Behavior Analysis and Violence Detection in Surveillance Videos traffic detection system